WEEK 5 - Payment Transaction Management System

Create Payment Table

Payment Entity

```
SQL> CREATE TABLE Payment (    Payment_ID NUMBER PRIMARY KEY,    Order_ID NUMBER,    Payment_Date DATE,    Payment_Mode VARCHAR2(20),    Payment_Sta
tus VARCHAR2(20),    Transaction_Amount NUMBER(10,2));

Table created.
```

Store Payment Details

```
SQL> INSERT INTO Payment VALUES (1, 1001, DATE '2026-09-27', 'UPI', 'Success', 67500);

1 row created.

SQL> INSERT INTO Payment VALUES (2, 1002, DATE '2026-09-27', 'Card', 'Success', 25000);

1 row created.

SQL> INSERT INTO Payment VALUES (3, 1003, DATE '2026-09-28', 'Cash', 'Success', 1500);

1 row created.

SQL> INSERT INTO Payment VALUES (4, 1004, DATE '2026-09-28', 'Net Banking', 'Pending', 4200);

1 row created.

SQL> INSERT INTO Payment VALUES (5, 1005, DATE '2026-09-29', 'UPI', 'Success', 8000);

1 row created.

SQL> INSERT INTO Payment VALUES (6, 1001, DATE '2026-09-29', 'Card', 'Failed', 1000);

1 row created.

SQL> INSERT INTO Payment VALUES (7, 1002, DATE '2026-09-30', 'UPI', 'Pending', 5000);

1 row created.

SQL> INSERT INTO Payment VALUES (8, 1003, DATE '2026-09-30', 'Net Banking', 'Failed', 2000);

1 row created.
```

```
SQL> SELECT * FROM Payment;
PAYMENT_ID    ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_ST
TRANSACTION_AMOUNT
---------- ---------- --------- -------------------- ----------
         1       1001 27-SEP-26 UPI                  Success
     67500

         2       1002 27-SEP-26 Card                 Success
     25000

         3       1003 28-SEP-26 Cash                 Success
      1500

PAYMENT_ID    ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_ST
TRANSACTION_AMOUNT
---------- ---------- --------- -------------------- ----------
         4       1004 28-SEP-26 Net Banking          Pending
      4200

         5       1005 29-SEP-26 UPI                  Success
      8000

         6       1001 29-SEP-26 Card                 Failed
      1000

PAYMENT_ID    ORDER_ID PAYMENT_D PAYMENT_MODE          PAYMENT_ST
TRANSACTION_AMOUNT
---------- ---------- --------- -------------------- ----------
         7       1002 30-SEP-26 UPI                  Pending
      5000

         8       1003 30-SEP-26 Net Banking          Failed
      2000

8 rows selected.
```

Constraints to the Payment Table

Foreign Key Constraint:

```
SQL> ALTER TABLE Payment ADD CONSTRAINT FK_Payment_Order FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID);

Table altered.
```

NOT NULL Constraint

```
SQL> ALTER TABLE Payment MODIFY Payment_Mode NOT NULL;
Table altered.
SQL> ALTER TABLE Payment MODIFY Payment_Status NOT NULL;
Table altered.
```

CHECK Constraint

```
SQL> ALTER TABLE Payment ADD CONSTRAINT CHK_Transaction_Amount CHECK (Transaction_Amount > 0);

Table altered.
```

Report 1: Payment Mode Analysis find;

* Number of transactions using UPI.

* Number of card payments.

* Most preferred payment method

```
SQL> SELECT COUNT(*) AS UPI_TRANSACTIONS FROM Payment WHERE Payment_Mode = 'UPI';

UPI_TRANSACTIONS
----------------
               3
```

```
SQL> SELECT COUNT(*) AS CARD_PAYMENTS FROM Payment WHERE Payment_Mode IN ('Credit Card', 'Debit Card', 'Card');

CARD_PAYMENTS
-------------
           2
```

```
SQL> SELECT Payment_Mode, COUNT(*) AS TRANSACTION_COUNT FROM Payment GROUP BY Payment_Mode ORDER BY TRANSACTION_C
OUNT DESC FETCH FIRST 1 ROW ONLY;

PAYMENT_MODE         TRANSACTION_COUNT
-------------------- -----------------
UPI                                  3
```

Report 2: Revenue Analysis finds;

* Total revenue generated.

* Revenue by payment method.

*Average transaction amount.

```
SQL> SELECT SUM(Transaction_Amount) AS Total_Revenue FROM Payment WHERE Payment_Status = 'Success';

TOTAL_REVENUE
-------------
       102000
```

```
SQL> SELECT Payment_Mode, SUM(Transaction_Amount) AS Total_Revenue FROM Payment WHERE Payment_Status = 'Success'
GROUP BY Payment_Mode ORDER BY Total_Revenue DESC;

PAYMENT_MODE         TOTAL_REVENUE
-------------------- -------------
UPI                          75500
Card                         25000
Cash                          1500
```

```
SQL> SELECT ROUND(AVG(Transaction_Amount), 2) AS Average_Transaction_Amount FROM Payment WHERE Payment_Status = '
Success';

AVERAGE_TRANSACTION_AMOUNT
--------------------------
                     25500
```

Report 3: Customer Payment History Display:

*Customer name

*Order ID

*Payment mode

*Amount

*Payment status

```
SQL> SELECT C.Customer_Name, O.Order_ID, P.Payment_Mode, P.Transaction_Amount AS Amount, P.Payment_Status FROM Customer C JO
IN Orders O ON C.Customer_ID = O.Customer_ID JOIN Payment P ON O.Order_ID = P.Order_ID ORDER BY P.Payment_Date;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PAYMENT_MODE            AMOUNT PAYMENT_ST
-------------------- ---------- ----------
Arun                                                     1001
UPI                      67500 Success

Priya                                                    1002
Card                     25000 Success

Kumar                                                    1003
Cash                      1500 Success


CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ----------
PAYMENT_MODE            AMOUNT PAYMENT_ST
-------------------- ---------- ----------
Divya                                                    1004
Net Banking               4200 Pending

Rahul                                                    1005
UPI                       8000 Success

Arun                                                     1001
Card                      1000 Failed
```

Payment analysis:

```
SQL> SELECT Payment_Mode, COUNT(*) AS Transaction_Count, SUM(Transaction_Amount) AS Total_Revenue, ROUND(AVG(Transaction_Amo
unt), 2) AS Avg_Transaction FROM Payment WHERE Payment_Status = 'Success' GROUP BY Payment_Mode ORDER BY Total_Revenue DESC;


PAYMENT_MODE         TRANSACTION_COUNT TOTAL_REVENUE AVG_TRANSACTION
-------------------- ----------------- ------------- ---------------
UPI                                  2         75500           37750
Card                                 1         25000           25000
Cash                                 1          1500            1500
```